

Phase	Focus	Main work	Deliverable
FP4	Architecture & Requirements	Define the final web-app architecture, separate UI/audio/TTS logic, establish supported browsers/OSes, define application requirements and acceptance criteria, organize the codebase.	Final architecture + project structure
FP5	Audio Pipeline & Cross-Platform Input	Make microphone recording and uploaded audio reliable across Windows/macOS and major browsers. Normalize MP3, M4A, WAV, FLAC, etc. Validate recordings and handle invalid files.	Reliable cross-platform audio pipeline
FP6	TTS & Voice-Cloning Engine	Refactor XTTS into a dedicated backend service/module. Harden model loading, CPU/GPU handling, language validation, generation, error handling, and temporary-file management. Verify languages.	Stable TTS/voice-cloning backend
FP7	Web UI/UX	Turn the prototype into a polished interface: clear workflow, recording/upload preview, language selection, generation status, audio comparison, playback/download controls, validation and user-friendly errors.	Feature-complete web interface
FP8	Responsible Use, Privacy & Security	Implement consent/acceptable-use messaging, define audio retention/deletion behavior, document local/server processing, protect uploaded files, review authentication/access requirements, and document voice-cloning risks and model licensing.	Responsible-use + security/privacy implementation
FP9	Testing & Quality Assurance	Build unit and integration tests. Test the complete browser workflow, audio formats, languages, error cases, Windows/macOS browsers, CPU/GPU configurations, and concurrent/use-recovery scenarios as appropriate.	Test suite + QA results

FP10	Web Deployment & Production Setup	Choose hosting/deployment architecture, configure the application server, model storage/download strategy, environment variables, logging, resource limits, HTTPS, domain/access configuration, and deployment process. Test a clean deployment from scratch.	Deployed production candidate
FP11	Final Hardening & Capstone Release	Perform end-to-end production testing, fix remaining defects, optimize performance, clean code, finalize documentation, prepare demo scenarios/screenshots, and freeze the release.	Production-ready v1.0 + final capstone materials